

Lab 05 –

While loop and User Defined Functions

Programming Exercise:

Q.1 Write down a Python program, using While loop that generates Odd no's in between 1 to 100.

Source Code:

```
1
2     number = 1
3     while number <= 100:
4         print(number, end=' '); number += 2
5
```

Output:

C:\Users\hp\PycharmProjects\main\venv\Scripts\python.exe "C:\Users\hp\PycharmProjects\main\main 3.py"

1 3 5 7 9 11 13 15 17 19 21 23 25 27 29 31 33 35 37 39 41 43 45 47 49 51 53 55 57 59 61 63 65 67 69 71 73 75 77 79 81 83 85 87 89 91 93 95 97 99

Process finished with exit code 0

Q.2 Write down a Python Program using While loop to generate the following outputs

a)

```
*****
*****
*****
*****

*****
*****
*****
*****
```

Source Code:

```
1 rows = 8 # Total number of rows to print
2 count = 1
3
4 while count <= rows:
5     if count <= 4:
6         print(" " * 8 + "*" * 14) # First 4 rows with 8 leading spaces
7     else:
8         print(" " * 24 + "*" * 14) # Last 4 rows with 24 leading spaces
9     count += 1
```

Output:

```
C:\Users\hp\PycharmProjects\main\env\Script
```

```
*****
*****
*****
*****

*****
*****
*****
*****
```

```
Process finished with exit code 0
```

b)

```
*
**
***
****
*****
*****
*****
*****
*****
```

Source Code:

```
1 rows = 8 # Total number of rows to print
2 count = 1
3
4 while count <= rows:
5     print("*" * count) # Print `count` number of stars in each row
6     count += 1
```

Output:

C:\Users\hp\PycharmProjects\main\ver

```
*
**
***
****
*****
*****
*****
*****
*****
```

Process finished with exit code 0

c)

```
*****
*****
*****
****
***
**
*
```

Source Code:

```
1 rows = 7 # Total number of rows to print
2 count = rows
3
4 while count >= 1:
5     print("*" * count) # Print `count` number of stars in each row
6     count -= 1
```

Output:

C:\Users\hp\PycharmProjects\main\venv\Sc

```
*****
*****
*****
****
***
**
*
```

Process finished with exit code 0

Q.3 Write down a python program having one function for calculating factorial of a no. And call that function within a While loop to generate factorial of numbers from 0 to 10

Source Code:

```
1  # Function to calculate the factorial of a number
2  def factorial(n): 1 usage
3      fact = 1
4      while n > 1:
5          fact *= n
6          n -= 1
7      return fact
8
9  # Using a while loop to calculate factorials of numbers from 0 to 10
10 num = 0
11 while num <= 10:
12     print(f"Factorial of {num} is {factorial(num)}")
13     num += 1
14
```

Output:

```
C:\Users\hp\PycharmProjects\main\ver
```

```
Factorial of 0 is 1
```

```
Factorial of 1 is 1
```

```
Factorial of 2 is 2
```

```
Factorial of 3 is 6
```

```
Factorial of 4 is 24
```

```
Factorial of 5 is 120
```

```
Factorial of 6 is 720
```

```
Factorial of 7 is 5040
```

```
Factorial of 8 is 40320
```

```
Factorial of 9 is 362880
```

```
Factorial of 10 is 3628800
```

```
Process finished with exit code 0
```

